

Group:

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Title:

"Assessing the Vulnerability of Modern Routers to ARP Spoofing Attacks and Exploring Potential Countermeasures"

Goal and Motivation:

The primary goal of this project is to investigate whether modern routers are still susceptible to ARP (Address Resolution Protocol) spoofing attacks, despite advancements in network security. ARP spoofing remains a significant threat as it can facilitate Man-in-the-Middle (MITM) attacks, leading to data breaches, credential theft, and other malicious activities. By testing a range of modern routers from different manufacturers, we aim to identify vulnerabilities and understand the underlying reasons for their susceptibility or resistance to ARP spoofing.

This project is worthwhile because ARP spoofing attacks continue to pose a serious risk to network security, especially in industrial and organizational settings where legacy systems are still in use. By identifying vulnerabilities and exploring potential countermeasures, we hope to contribute to the development of more secure networking technologies and raise awareness about the ongoing risks associated with ARP spoofing.

Related Work:

ARP spoofing has been a well-known vulnerability in network security for decades. Despite the development of various countermeasures, it remains a persistent threat due to the inherent design limitations of the ARP protocol, which lacks authentication mechanisms.

One of the most notable defenses against ARP spoofing is Dynamic ARP Inspection (DAI), a security feature implemented in many modern switches. DAI works by validating ARP packets in a network to ensure that they are not spoofed. It uses a combination of DHCP snooping and ARP inspection to build a trusted database of IP-to-MAC address bindings. When an ARP packet is received, DAI checks it against this database. If the packet does not match the trusted bindings, it is dropped, preventing ARP spoofing attacks. However, DAI is not universally deployed, and its effectiveness depends on proper configuration and the use of DHCP snooping, which may not be feasible in all network environments.

Other related defenses include Secure ARP (S-ARP), which introduces cryptographic authentication to ARP messages, and static ARP entries, which manually map IP addresses to MAC addresses. However, these solutions have limitations, such as scalability issues, performance overhead, and incompatibility with legacy systems. Additionally, the Zeus Trojan has demonstrated how ARP spoofing can be used in advanced attacks to intercept and manipulate network traffic, highlighting the need for more robust defenses.

Expected Contributions:

This project aims to make several novel contributions to the field of network security:

- **Novel Problem Identification:** By testing modern routers, we will determine whether ARP spoofing vulnerabilities still exist in current hardware, despite the availability of defenses like DAI and S-ARP.
- **Novel Solution Exploration:** We plan to explore potential countermeasures that could be applied to vulnerable routers, such as modifying firmware or implementing additional security protocols.
- **Novel Results:** Our findings will provide a comprehensive assessment of the current state of ARP spoofing vulnerabilities in modern routers, offering insights into why certain routers remain susceptible and how they can be hardened against such attacks.

Evaluation:

To evaluate the vulnerability of modern routers to ARP spoofing, we will:

- **Manual Testing:** Conduct hands-on testing of routers from various manufacturers (e.g., Tenda, Netgear) to determine their susceptibility to ARP spoofing using some open source frameworks.
- **Framework Analysis:** Examine the underlying firmware and software frameworks of these routers to understand why they are vulnerable or resistant to ARP spoofing.
- **Benchmarking:** Use existing tools and techniques (e.g., Ettercap, Wireshark) to simulate ARP spoofing attacks and measure the effectiveness of potential countermeasures.

Results:

By the end of the quarter, we plan to present the following results:

- **Vulnerability Assessment:** A detailed report on the susceptibility of tested routers to ARP spoofing, including which models are vulnerable and which are not.
- **Framework Insights:** An analysis of the underlying reasons for the vulnerabilities or resistance of specific routers, focusing on their firmware and software design.
- **Countermeasure Proposals:** Recommendations for hardening vulnerable routers against ARP spoofing, including potential modifications to firmware or network configurations.
- **Demonstration (optional):** A live demonstration of ARP spoofing on a vulnerable router and the effectiveness of proposed countermeasures.

Progress So Far:

We have made some progress in our project by successfully demonstrating the existence of ARP spoofing vulnerabilities in modern routers. Specifically, we have conducted ARP spoofing

attacks on two widely used home routers: the *Tenda RX2 Pro* (November 2022 Release) and the *Netgear R6700v3* (2014 release). These experiments confirm that ARP spoofing remains a viable threat even in contemporary networking hardware.

- **Tenda RX2 Pro:** We were able to successfully execute an ARP spoofing attack on this router, demonstrating how an attacker could intercept and manipulate network traffic. This router, which is commonly used in home networks, showed clear vulnerabilities to ARP spoofing, highlighting the need for improved security measures in consumer-grade devices.
- **Netgear R6700v3:** Despite being a relatively older model (released in 2014), this router is still in use in many home and small office environments. Our successful ARP spoofing attack on this device further underscores the persistence of this vulnerability, even in routers that have been on the market for several years.

These initial experiments have provided us with valuable insights into the current state of ARP spoofing vulnerabilities in home routers. However, our work is not limited to home routers. We are expanding our testing to include several other home routers from different manufacturers to gather a more comprehensive dataset on the prevalence of ARP spoofing vulnerabilities in modern devices.

Additionally, we are planning to extend our experiments to **enterprise-level** routers (possibly TP-link for starters). Enterprise routers typically come with more advanced security features, such as **Dynamic ARP Inspection (DAI)** and other protective mechanisms. By testing these routers, we aim to evaluate the effectiveness of these security measures in preventing ARP spoofing attacks. This will allow us to compare the security posture of consumer-grade routers with that of enterprise-grade devices, providing a more holistic understanding of the current landscape of ARP spoofing vulnerabilities.